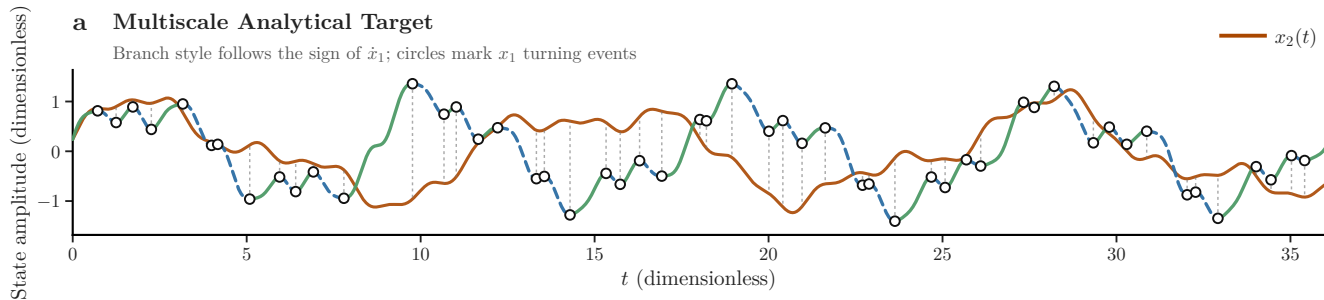


Relational Coordinates Can Simplify the Organization of a Scientific Target

— $\dot{x}_1 > 0$ - - $\dot{x}_1 < 0$ ○ x_1 turning event

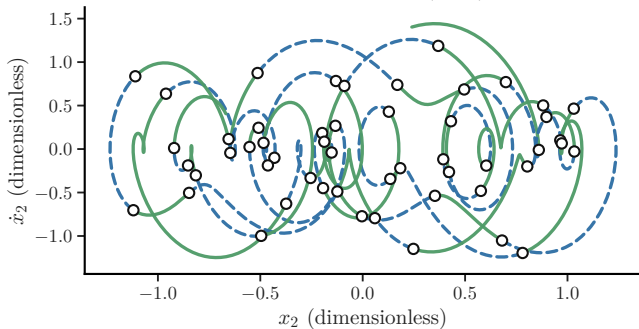
a Multiscale Analytical Target

Branch style follows the sign of $\dot{x}_1$; circles mark x_1 turning events



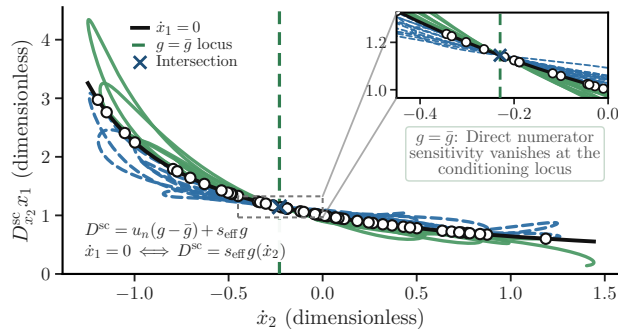
b Level-Rate Coordinates

The target branches remain folded in $(x_2, \dot{x}_2)$



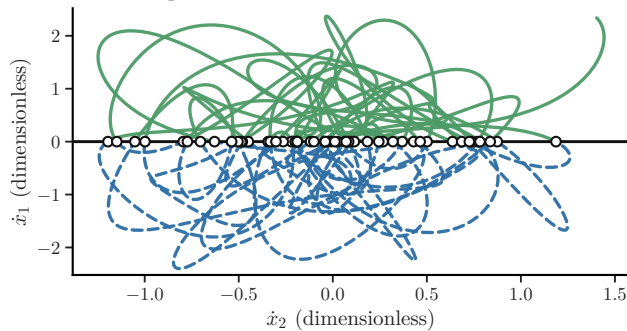
c Sensitivity-Centered Phase Coordinates

Manifold geometry and conditioning organize the target



d Rate-Rate Coordinates

Turning events lie on $\dot{x}_1 = 0$ with reference-rate context



e Rate-Acceleration Coordinates

Turning events map to local curvature values on $\dot{x}_1 = 0$

